

Vectors and Geometry

1.1 Introduction

This chapter covers the fundamental concepts of linear algebra related to vectors and geometry. We will define vectors, explore algebraic operations, study vector products, and apply these concepts to the vector representation of lines and planes.

1.2 Definition of a Vector

A vector is defined as any entity which possesses both **magnitude** and **direction**. This stands in contrast to a **scalar**, which is a quantity defined only by magnitude (e.g., mass, temperature).

Example 1.2.1. • **Force:** A pushing force of 5 Newtons at an angle of 30° is a vector because it has magnitude (5N) and direction.

- **Velocity:** Wind blowing at 3 m/s towards the Northwest is a vector.

1.2.1 Coordinate Representation

In a standard coordinate system (e.g., the xy -plane), a vector $\mathbf{v}$ can be represented by its components. If a vector has a magnitude $|\mathbf{v}|$ and makes an angle θ with the horizontal axis, its coordinates (V_x, V_y) are given by:

$$V_x = |\mathbf{v}| \cos \theta \quad (\text{x-component}) \quad (1.1)$$

$$V_y = |\mathbf{v}| \sin \theta \quad (\text{y-component}) \quad (1.2)$$

Conversely, if given the components (V_x, V_y) , we can determine the magnitude and direction:

- **Magnitude:** Using the Pythagorean theorem, $|\mathbf{v}| = \sqrt{V_x^2 + V_y^2}$.
- **Direction:** The angle is found using $\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$.

In computer science, vectors often extend beyond 2 or 3 dimensions to n -dimensions (e.g., a vector representing the incomes of a population), though visualization is difficult for $n > 3$.

1.3 Algebraic Operations on Vectors

1.3.1 Addition and Subtraction

Vector addition is performed element-wise. If $\mathbf{u} = (u_x, u_y)$ and $\mathbf{v} = (v_x, v_y)$, their sum is:

$$\mathbf{w} = \mathbf{u} + \mathbf{v} = (u_x + v_x, u_y + v_y)$$

Note 1.3.1. Parallelogram Rule: Geometrically, the sum of two vectors $\mathbf{u}$ and $\mathbf{v}$ corresponds to the diagonal of the parallelogram formed by these vectors originating from the same point.

Subtraction is defined as the addition of a negative vector: $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

1.3.2 Scalar Multiplication (Scaling)

Multiplying a vector by a scalar λ scales the magnitude of the vector without changing its line of action.

- If $|\lambda| > 1$, the vector is lengthened (stretched).
- If $|\lambda| < 1$, the vector is shortened (shrunk).
- If $\lambda < 0$, the direction of the vector is reversed.

1.4 Unit Vectors and Basis

1.4.1 Unit Vectors

A **unit vector** is any vector with a magnitude of exactly 1. Any non-zero vector $\mathbf{v}$ can be converted into a unit vector $\hat{\mathbf{v}}$ pointing in the same direction by dividing by its magnitude:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

1.4.2 Basis Vectors

Definition 1.4.1. Vectors are **linearly independent** if none of them can be written as a linear combination of the others. The equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

has only the trivial solution $c_1 = c_2 = \cdots = c_n = 0$.

Definition 1.4.2. A set of vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is called a **basis** if it satisfies two conditions:

1. **Linear Independence:** The vectors are linearly independent.
2. **Spanning the Space:** The vectors span V . For every vector $\mathbf{w} \in V$, there exist scalars $a_1, a_2, \dots, a_n$ such that:

$$\mathbf{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$$

Basis vectors act as building blocks for the space. In the standard Cartesian plane ($\mathbb{R}^2$), the standard basis vectors are:

- $\hat{i} = (1, 0)$ (along the x-axis)
- $\hat{j} = (0, 1)$ (along the y-axis)

Any vector in the plane can be expressed as a **linear combination** of these basis vectors: $\mathbf{v} = a\hat{i} + b\hat{j}$. The standard basis vectors for $\mathbb{R}^3$ are:

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Any vector $\mathbf{v} = (x, y, z)$ can be written uniquely as $\mathbf{v} = x\hat{i} + y\hat{j} + z\hat{k}$.

1.5 Products on Vectors

There are two primary ways to multiply vectors: the dot product and the cross product.

1.5.1 The Dot Product (Scalar Product)

The dot product results in a scalar value. For two vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n \quad (1.3)$$

Alternatively, geometrically:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos \theta \quad (1.4)$$

where θ is the angle between the vectors. This relationship allows us to calculate the angle between vectors.

- If $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are **orthogonal** (90°).
- The dot product can be viewed as the magnitude of $\mathbf{u}$ multiplied by the projection of $\mathbf{v}$ onto $\mathbf{u}$.

1.5.2 The Cross Product (Vector Product)

The cross product results in a vector that is orthogonal to both input vectors.

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}||\mathbf{b}| \sin \theta) \hat{n} \quad (1.5)$$

where $\hat{n}$ is the unit normal vector determined by the right-hand rule.

- The magnitude $|\mathbf{a} \times \mathbf{b}|$ equals the **area of the parallelogram** spanned by $\mathbf{a}$ and $\mathbf{b}$.
- If $\mathbf{a} \times \mathbf{b} = \mathbf{0}$, the vectors are parallel.

1.6 Vector Representation of Lines and Planes

1.6.1 Lines in Space

A line can be represented vectorially using a starting position vector $\mathbf{r}_0$ and a direction vector $\mathbf{v}$. The equation for any point $\mathbf{r}$ on the line is:

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v} \quad (1.6)$$

where t is a scalar parameter. To find $\mathbf{v}$, one can subtract position vectors of two points

on the line: $\mathbf{v} = \mathbf{r}_1 - \mathbf{r}_0 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} - \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}.$

1.6.2 Planes in Space

A plane can be defined by a point $\mathbf{r}_0$ and two non-parallel direction vectors $\mathbf{v}$ and $\mathbf{u}$ lying in the plane. The equation is:

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v} + s\mathbf{u} \quad (1.7)$$

To find $\mathbf{v}$, one can subtract position vectors: $\mathbf{v} = \mathbf{r}_1 - \mathbf{r}_0 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} - \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$ and to find $\mathbf{u}$,

one can subtract position vectors: $\mathbf{u} = \mathbf{r}_2 - \mathbf{r}_0 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} - \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}.$

Alternatively, a plane is defined by a point P_0 (position vector $\mathbf{r}_0$) and a **normal vector** $\mathbf{n}$. Any point P (position vector $\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$) in the plane satisfies the condition that the vector $\vec{P_0P}$ is orthogonal to $\mathbf{n}$:

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

This leads to the Cartesian equation of the plane $Ax + By + Cz = K$.

If the vector equation of a line is given as $\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}$, $\mathbf{u}$ and $\mathbf{v}$ are direction vectors, then the normal will be in the direction of the cross product of direction vectors that lie in the plane.

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}$$

We can then use

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

to get the Cartesian equation.